

Advances in Finite Element Methods and Parallel Scientific Computing

November 28 – December 2, 2026
IIT Bhubaneswar

Academic Program

Venue: Room No. L02M, Lalitgiri Hall of Learning

Inauguration Program: 9:00 AM – 9:30 AM, L02M (Lalitgiri Hall of Learning), 28th November 2026

TBDS: To Be Decided Soon

Day 1: November 28, 2026

Time	Speaker / Title of the Talk
	Chair: Prof. Bhupen Deka
9:30 AM – 10:20 AM	Dr. Asha Dond <i>Lecture Series on Finite Element Methods for PDEs – Lecture 1</i>
10:25 AM – 11:05 AM	Prof. Rajen Sinha <i>TBDS</i>
11:05 AM – 11:35 AM: TEA BREAK	
11:35 AM – 12:25 PM	Dr. Arbaz Khan <i>Lecture Series on FEM for Multiphysics Problems – Lecture 1</i>
12:30 PM – 12:50 PM	Contributed Talk: Rohit Kumar Yadav <i>A parameter uniform Numerical method for Singularly Perturbed Semilinear Convection diffusion problems with discontinuous source term</i>
1:00 PM – 2:30 PM: LUNCH BREAK	
	Chair: Dr. Abhinav Jha
2:30 PM – 3:10 PM	Dr. Rakesh Kumar <i>TBDS</i>
3:15 PM – 3:35 PM	Contributed Talk: Ajay Kumar <i>Hybrid high-order methods for sixth order problems with a mixed formulation</i>
3:40 PM – 4:00 PM	Contributed Talk: Prince Bharti <i>Analysis of a hybrid high-order method for on a parabolic convection diffusion integro-differential equations on general meshes</i>
4:00 PM – 4:25 PM: TEA BREAK	
4:25 PM – 4:45 PM	Contributed Talk: Dilip Sarkar <i>A Finite Element Framework for Domain Decomposition with Non-Matching Grids: A Nitsche-Based Approach</i>
4:50 PM – 5:10 PM	Contributed Talk: Kumar Mritunjay

Day 2: November 29, 2026

Time	Speaker / Title of the Talk
	Chair: Dr. Bankim Chandra Mandal
9:30 AM – 10:20 AM	Dr. Asha Dond <i>Lecture Series on Finite Element Methods for PDEs – Lecture 2</i>
10:25 AM – 11:05 AM	Dr. Saumya Bajpai <i>TBDS</i>
11:05 AM – 11:35 AM: TEA BREAK	
11:35 AM – 12:15 PM	Dr. Pratibhamoy Das <i>TBDS</i>
12:20 PM – 1:00 PM	Dr. Y. Bhumkar <i>TBDS</i>
1:00 PM – 2:30 PM: LUNCH BREAK	
	Chair: Dr. Sangita Yadav
2:30 PM – 2:50 PM	Contributed Talk: Suraj Kumar <i>Parameter-Uniform Convergence and Superconvergence of NIPG Method for Degenerate Singularly Perturbed Convection-Diffusion Problems</i>
2:50 PM – 3:10 PM	Contributed Talk: Jitendra Nath Naik <i>Optimal Error Estimates of a Finite Element Method for Semilinear SPDEs with Non-smooth Initial Data</i>
3:15 PM – 3:55 PM	Dr. Gouranga Mullick <i>TBDS</i>
3:55 PM – 4:20 PM: TEA BREAK	
4:20 PM – 4:40 PM	Contributed Talk: Arijit Pal <i>Dimensional-Splitting Weak Galerkin FEM for Singularly Perturbed Initial-Boundary Value Problems</i>
4:40 PM – 5:00 PM	Contributed Talk: Dr. Rashmita Baruah <i>A Weak Galerkin Method with Forward Euler Time Discretization for Singularly Perturbed Parabolic Interface Problems</i>
5:00 PM – 5:20 PM	Contributed Talk: Pratyay Mondal <i>A discontinuous Galerkin pressure correction scheme for the Oldroyd model of order one</i>

Day 3: November 30, 2026

Time	Speaker / Title of the Talk
	Chair: Prof. Rajen Sinha
9:30 AM – 10:20 AM	Dr. Arbaz Khan <i>Lecture Series on FEM for Multiphysics Problems – Lecture 2</i>
10:25 AM – 11:05 AM	Prof. Rajendra K. Ray <i>TBDS</i>
11:05 AM – 11:25 AM: TEA BREAK	
11:25 AM – 12:15 PM	Prof. Bhupen Deka <i>Lecture series - 1</i>
12:20 PM – 1:00 PM	Dr. Sangita Yadav <i>TBDS</i>
1:00 PM – 2:30 PM: LUNCH BREAK	
3:00 PM – 7:30 PM	Free Time/Local Excursion

Day 4: December 1, 2026

Time	Speaker / Title of the Talk
	Chair: Prof. Rajendra K. Ray
9:30 AM – 10:20 AM	Dr. Abhinav Jha <i>Lecture Series – Lecture 1</i>
10:25 AM – 11:05 AM	Prof. Neela Natarajan <i>TBDS</i>
11:05 AM – 11:15 AM: TEA BREAK	
11:20 AM – 12:00 PM	Dr. Nagaiah Chamakuri <i>TBDS</i>
12:05 PM – 12:25 PM	Contributed Talk: Deeksha Tomer <i>Domain Decomposition Methods for PDEs with Time Delay</i>
12:25 PM – 1:10 PM	Dr. Surya Maharana <i>TBDS</i>
1:10 PM – 2:30 PM: LUNCH BREAK	
	Chair: Dr. Sanjib Acharya
2:30 PM – 3:20 PM	Prof. Bhupen Deka <i>Lecture series - 2</i>
3:25 PM – 3:45 PM	Contributed Talk: Monirul Islam <i>Unconditionally Energy Stable Ultra-Weak Discontinuous Galerkin Scheme with Generalized Numerical Fluxes for the Allen-Cahn Equation</i>
3:50 PM – 4:10 PM	Contributed Talk: Nishant Ranwan <i>A priori error analysis for the loosely coupled Robin-Robin partitioned scheme for a fully discrete fluid-structure interaction problem</i>
4:10 PM – 4:30 PM: TEA BREAK	
4:30 PM – 4:50 PM	Contributed Talk: Ashish Kumar Saini <i>Enhancing thermal performance in square enclosures filled with oxide nano-fluids subjected to magnetic fields: a numerical study</i>
4:55 PM – 5:15 PM	Contributed Talk: Dr. Harish Mallesham Nagula <i>Vectorized Mortar Primal Hybrid Finite Element Method</i>

Day 5: December 2, 2026

Time	Speaker / Title of the Talk
	Chair: Dr. Nagaiah Chamakuri
9:30 AM – 10:20 AM	Dr. Abhinav Jha <i>Lecture Series – Lecture 2</i>
10:25 AM – 11:05 AM	Prof. Sarvesh Kumar <i>TBDS</i>
11:05 AM – 11:35 AM: TEA BREAK	
11:35 AM – 12:15 PM	Contributed Talk: Dr. Achyuta Ranjan Dutta Mohapatra <i>Energy norm error estimates of a hybrid high-order method for the linear parabolic integro-differential equations on general meshes</i>
12:20 PM – 1:00 PM	Contributed Talk: Anupama Ghorai <i>A high-order finite element method for multi-dimensional population balance equations.</i>
1:00 PM – 3:00 PM: SOCIAL LUNCH	
	Chair: Dr. Arbaz Khan
3:00 PM – 3:20 PM	Contributed Talk: Dr. Ruma Rani Maity <i>An Energy-Decreasing Algorithm for the Finite Element Approximation of Ferronematic Equilibrium States</i>
3:25 PM – 3:45 PM	Contributed Talk: Dr. Shubhangi Dwivedi <i>Managing Connected Ecological Patches Under Species Dispersal Using Optimal Control</i>
3:50 PM – 4:30 PM: CLOSING CEREMONY	